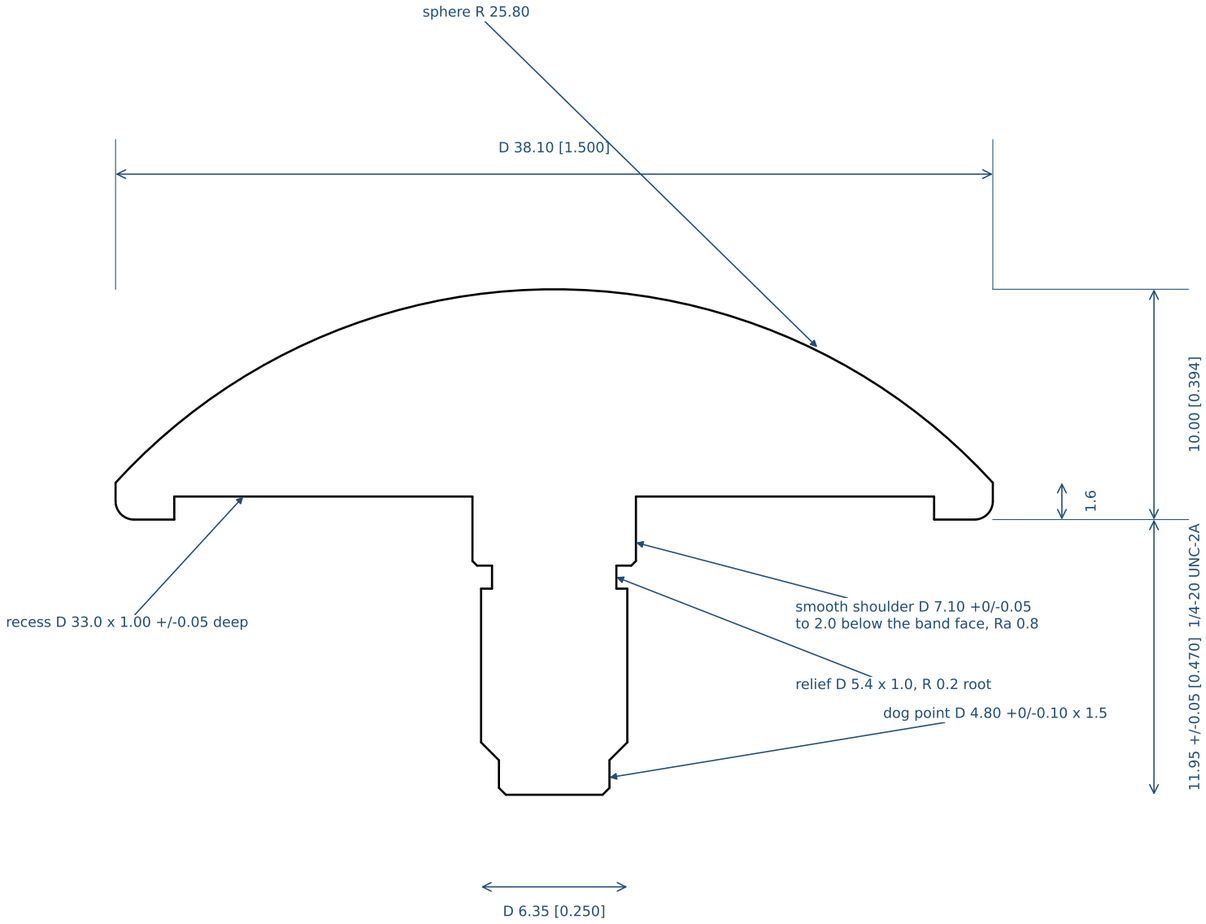
SECTION THROUGH THE AXIS



TOP VIEW

NOTES:

1. Dome is a sphere segment, R 25.80, height 8.4 above a 1.6 straight band; base edge R 0.8. NO knurl, no tool marks on the dome.
2. Thread 1/4-20 UNC-2A (class BEFORE plating), full form from the run-in to the relief groove D 5.4 x 1.0 (R 0.2 at the groove root). TIP: unthreaded dog point D 4.80 +0/-0.10 x 1.5 long (must enter the insert's 4.98 MIN minor dia), 0.3 x 45 chamfer, 45 deg run-in to the thread.
3. Stud length 11.95 +/-0.05 from the band (underside) face to the tip. Stud axis to dome axis concentric within 0.05. Underside face flat and square to the stud within 0.05.
4. Smooth shoulder D 7.10 +0/-0.05 from the recess floor to 2.0 below the band face, Ra 0.8, 0.2 lead-in: the record hangs and centres on it (spindle hole 7.24 min). Underside recess D 33.0 x 1.00 +/-0.05 deep, flat floor.
5. Plating: rack on the STUD only. MASK the thread and the shoulder (do NOT chase after plating: chrome flakes). Any nickel on the thread 5 um max; the thread must accept a 1/4-20 2B GO gauge AFTER plating. Dome and band are Class A: inspect under daylight at 300 mm, zero pits/haze.
6. We fit after plating: felt washer 31.75 OD x 9.5 ID x 1.6 in the recess (0.6 proud free, 0.3 at 50 N) and a loose clear PET label washer 37.5 x 7.5 x 0.25 under the cap. REV D (2026-11-11): stud 11.95 (rev C 11.70), shoulder added, tolerances added, separate-stud option deleted (a tapped D 7.10 shoulder would have a 0.37 wall).
7. Finish variants share the same part number with suffix -CR (chrome), -BR (brass), -BK (black).

RECORD SCONCE | DOME CAP 1.5 in, 12 in record

PART NO: RS-CAP-12 REV: D DATE: 2026-11-11 SUPERSEDES: rev A, B, C (do not build or quote)

MATERIAL: Brass C360 (CDA 360) bar, turned in ONE piece with the 1/4-20 stud. No separate or bonded stud.

FINISH: CHROME: polish to mirror (Ra <= 0.1 um) then copper + bright nickel + decorative chrome per ASTM B456 SC2 (or PVD chrome over bright nickel). BRASS: polish + clear lacquer or brushed + lacquer. BLACK: black chrome over bright nickel, or PVD black (TiC/DLC), or black nickel + lacquer

UNITS: mm [in] THIRD-ANGLE PROJECTION SCALE: as shown

GENERAL TOLERANCES: x.x +/-0.3 x.xx +/-0.1 angles +/-0.5 deg unless noted